MEDIUM ALGEBRA

Systems of two linear equations in two variables

02

10 questions · ~15 min

SAT QUESTION BANK 2026-05-29

Q1 ALGEBRA

$$\begin{aligned} y &= 2x - 3 \\ 3y &= 5x \end{aligned}$$

In the solution to the system of equations above, what is the value of y ?

A. -15

B. -9

C. 9

D. 15

Q2 ALGEBRA

$$y = 6x + 16$$

$$-7x - y = 36$$

What is the solution x, y to the given system of equations?

A. $-4, -8$

B. $-\frac{20}{13}, -\frac{80}{13}$

C. $4, 40$

D. $20, 136$

Q3 ALGEBRA

A proposal for a new library was included on an election ballot. A radio show stated that 3 times as many people voted in favor of the proposal as people who voted against it. A social media post reported that 15,000 more people voted in favor of the proposal than voted against it. Based on these data, how many people voted against the proposal?

A. 7,500

B. 15,000

C. 22,500

D. 45,000

Q4

ALGEBRA

$$y = 2x + 10$$

$$y = 2x - 1$$

At how many points do the graphs of the given equations intersect in the *xy*-plane?

A. Zero

B. Exactly one

C. Exactly two

D. Infinitely many

Q5 GRID-IN ALGEBRA

$$2a + 8b = 198$$

$$2a + 4b = 98$$

The solution to the given system of equations is a, b. What is the value of b?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q6 ALGEBRA

The sum of a number x and 7 is twice as large as a number y. The number y is 3 less than the number x. Which system of equations describes this situation?

A. $x + 7 = 2y$
$y = x - 3$

B. $x + 7 = 2y$
$y = 3 - x$

C. $2x + 7 = y$
$y = x - 3$

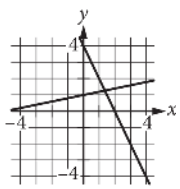
D. $2x + 7 = y$
$y = 3 - x$

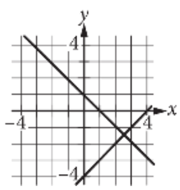
Q7

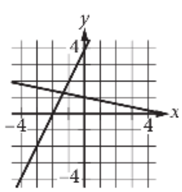
ALGEBRA

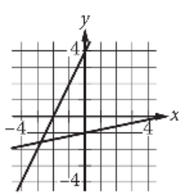
$$x + 5y = 5$$
$$2x - y = -4$$

Which of the following graphs in the xy-plane could be used to solve the system of equations above?

A. 

B. 

C. 

D. 

Q8 ALGEBRA

A group of 202 people went on an overnight camping trip, taking 60 tents with them. Some of the tents held 2 people each, and the rest held 4 people each. Assuming all the tents were filled to capacity and every person got to sleep in a tent, exactly how many of the tents were 2-person tents?

A. 30

B. 20

C. 19

D. 18

Q9 ALGEBRA

A bus traveled on the highway and on local roads to complete a trip of 160 miles. The trip took 4 hours. The bus traveled at an average speed of $55 \text{ miles per hour mph}$ on the highway and an average speed of 25 mph on local roads. If x is the time, in hours, the bus traveled on the highway and y is the time, in hours, it traveled on local roads, which system of equations represents this situation?

A. $55x + 25y = 4$
$x + y = 160$

B. $55x + 25y = 160$
$x + y = 4$

C. $25x + 55y = 4$
$x + y = 160$

D. $25x + 55y = 160$
$x + y = 4$

Q10 ALGEBRA

$$y = 6x + 3$$

One of the two equations in a system of linear equations is given. The system has infinitely many solutions. Which equation could be the second equation in this system?

A. $y = 2(6x) + 3$

B. $y = 2(6x + 3)$

C. $2(y) = 2(6x) + 3$

D. $2(y) = 2(6x + 3)$